

# Threading Micro-optimizations Report

Project G45 · Sprint 4 · Based on CPU profiling (cpu-simpleperf-G45.trace)

|            |                                                             |
|------------|-------------------------------------------------------------|
| Author     | juan-jerez                                                  |
| Date       | 2026-05-23                                                  |
| Tool       | Android Simpleperf — events: cpu-clock + sched:sched_switch |
| Trace file | cpu-simpleperf-G45.trace (93 MB)                            |
| Changes    | 3 ViewModels / Managers modified across 5 files             |

## 1. Analysis Methodology

The trace was recorded with Simpleperf using two simultaneous events: **cpu-clock** (CPU cycle samples every ~10 ms) and **sched:sched\_switch** (kernel-level thread context switches). The binary file was inspected with **strings** to extract the method symbols present in the sampled call stacks. Each appearance of a symbol indicates that method was **actively executing on the CPU** during a sample.

A baseline was established by observing that most ViewModels appear **2 times** in the symbol set (e.g. *AgendaViewModel.<init>*, *HomeViewModel.<init>*). Symbols exceeding that baseline were flagged as anomalies warranting investigation.

| Symbol                                    | CPU samples | Interpretation             |
|-------------------------------------------|-------------|----------------------------|
| AgendaViewModel.                          | 2×          | Normal baseline            |
| HomeViewModel.                            | 2×          | Normal baseline            |
| PqrListViewModel.                         | 6×          | ■ 3× above baseline        |
| TutorViewModel.                           | 6×          | ■ 3× above baseline        |
| PqrViewModelFactory.create                | 6×          | ■ 3× above baseline        |
| listenToUserSessions\$1.invoke            | 4×          | ■ concurrent with 2 others |
| listenToReviews\$1.invoke                 | 4×          | ■ concurrent with 2 others |
| listenToUnreadNovelties\$1.invoke         | 4×          | ■ concurrent with 2 others |
| flatMapLatest\$1.invoke<br>(ReviewListVM) | 4×          | ■ 2 restarts per reload    |

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## 2. Finding 1 — Concurrent Fetches from ViewModel Recreation

**Profiling evidence:** `PqrListViewModel.<init>` → **6 CPU samples** (3× the 2-sample baseline).  
`PqrViewModelFactory.create` → **6 CPU samples**. Other ViewModels measured at baseline: 2 samples.

### Diagnosis

A well-behaved ViewModel should appear **once in its constructor** during a profiling session. `PqrListViewModel` appears 6 times, indicating it was constructed at least 3 times during the session (each construction yields ~2 samples given the 10 ms cpu-clock interval). This happens because `PqrListScreen` is navigated to multiple times and each navigation instantiates a new ViewModel whose `init()` immediately fires `fetchPqrs()`.

Without a cancellation guard, if a previous ViewModel was not destroyed before a new one started, **both fetches run concurrently** inside their respective `viewModelScope` instances: two parallel Firestore reads for the same data, with two `_uiState` updates arriving in non-deterministic order.

### Modified file

[PqrListViewModel.kt](#) — added `fetchJob: Job?` that cancels the previous fetch before launching a new one.

| ■ Before                                                                                                                                                                                                                        | ■ After                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>fun fetchPqrs() {     val userId = AuthHolder.authRepo         .getCurrentUserId() ?: return     viewModelScope.launch {         _uiState.update { it.copy(isLoading = true) }         // Firestore fetch...     } }</pre> | <pre>private var fetchJob: Job? = null  fun fetchPqrs() {     val userId = AuthHolder.authRepo         .getCurrentUserId() ?: return     fetchJob?.cancel() // cancel in-flight fetch     fetchJob = viewModelScope.launch {         _uiState.update { it.copy(isLoading = true) }         // Firestore fetch...     } }</pre> |

**Effect:** If the ViewModel is recreated while a fetch is in flight, `fetchJob?.cancel()` terminates the previous coroutine before launching the new one. This eliminates the race condition where two Firestore calls update `_uiState` in non-deterministic order.

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## 3. Finding 2 — Double-Emit Causing Redundant flatMapLatest Restarts

### Profiling

### evidence:

`ReviewListViewModel.listenReviews$1.invokeSuspend$$inlined$flatMapLatest$1.invoke` → **4 CPU samples**. Each `flatMapLatest` invocation cancels and relaunches an inner coroutine, producing wasted work.

## Diagnosis

The `reLoadReviews()` function forced a re-emission of the upstream flow by assigning **null and then the original value** back to `_dayFilter`. Because `StateFlow` only suppresses *identical* consecutive values, the `null → value` sequence produces **two distinct emissions** from the `combine()` operator. Each emission cancels the inner coroutine of `flatMapLatest` and launches a new one, resulting in **2 coroutine restarts per manual reload**: the first is cancelled mid-execution and discarded.

With 2 reload events during the profiling session (initial filter + manual reload), the lambda appears **4 times** in the trace instead of the expected 2.

## Modified file

[ReviewListViewModel.kt](#) — replaced the double-assignment trick with a `MutableSharedFlow` refresh trigger. Each manual reload emits exactly once.

| ■ Before                                                                                                                                                                                                                                                                                                                                                              | ■ After                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>fun reLoadReviews() {     _uiState.update { it.copy(error = "") }     val current = _dayFilter.value     _dayFilter.value = null // emit 1 →     restarts flatMapLatest     _dayFilter.value = current // emit 2 →     restarts flatMapLatest }  // In listenReviews(): combine(_isTutor, _dayFilter) { t, d -&gt;     Pair(t, d) } .flatMapLatest { ... }</pre> | <pre>private val _refreshTrigger =     MutableSharedFlow(extraBufferCapacity =         1)  fun reLoadReviews() {     _uiState.update { it.copy(error = "") }     _refreshTrigger.tryEmit(Unit) //     exactly 1 emit }  // In listenReviews(): combine(_isTutor, _dayFilter,     _refreshTrigger.onStart { emit(Unit) } ) { t, d, _ -&gt; Pair(t, d) } .flatMapLatest { ... }</pre> |

**Effect:** Each call to `reLoadReviews()` triggers exactly **one** `flatMapLatest` invocation instead of two. Cancelled mid-flight coroutines are eliminated, along with the partial Firestore calls they produced.

## 4. Finding 3 — Three Firestore Listeners Executing Simultaneously

**Profiling evidence:** Three Firestore callback lambdas appear at the same frequency (4× each) within the same trace window: `listenToUserSessions$1.invoke`, `listenToReviews$1.invoke`, `listenToUnreadNovelties$1.invoke`.

## Diagnosis

All three listeners are implemented as *callbackFlow*. The Firestore SDK dispatches each snapshot on its own internal thread pool. When all three receive an update simultaneously — for example at session start, when Firestore delivers the initial snapshot for each collection — their callbacks run in parallel, each calling *trySend()* which resumes the collector coroutine in *viewModelScope*.

The default buffer of *callbackFlow* is *Channel.BUFFERED* (64 slots). If Firestore sends multiple rapid snapshots (e.g. during a write + confirmation cycle) while the collector is still processing the previous one, those snapshots **accumulate in the buffer** and each triggers a redundant *\_uiState* update with identical or near-identical data.

## Modified files

*.conflate()* was appended to the three *callbackFlow* definitions: [ReservationsFireStoreManager.kt](#), [ProfileListsFireStoreManager.kt](#) and [NoveltyFirestoreManager.kt](#).

| ■ Before                                                                                                                                                                                                                   | ■ After                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>fun listenToUserSessions(...) : Flow&gt; = callbackFlow { // query + addSnapshotListener... awaitClose { registration.remove() } } // Default: Channel.BUFFERED (64 slots) // → intermediate snapshots queue up</pre> | <pre>fun listenToUserSessions(...) : Flow&gt; = callbackFlow { // query + addSnapshotListener... awaitClose { registration.remove() } }.conflate() // Buffer: 1 slot (latest value only) // → intermediate snapshots discarded</pre> |

**Effect:** With *conflate()*, if the collector is busy processing a snapshot and two more arrive, only the **latest** is delivered. Intermediate states are discarded without processing. The final result is always correct (most recent Firestore state) with fewer CPU cycles and fewer redundant *\_uiState* updates.

## 5. Change Summary

| # | Finding                                                  | Trace symbol                                                                             | Samples | Modified file                                                                                    |
|---|----------------------------------------------------------|------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------|
| 1 | Concurrent fetches from ViewModel recreation             | PqrListViewModel. 6×<br>PqrViewModelFactory.create 6×                                    | 6×      | PqrListViewModel.kt                                                                              |
| 2 | Double-emit causing redundant flatMapLatest restarts     | listenReviews\$<br>flatMapLatest\$1.invoke 4×                                            | 4×      | ReviewListViewModel.kt                                                                           |
| 3 | 3 simultaneous Firestore listeners with buffered channel | listenToUserSessions\$1<br>listenToReviews\$1<br>listenToUnreadNovelties\$1<br>(4× each) | 4× each | ReservationsFireStoreManager.kt<br>ProfileListsFireStoreManager.kt<br>NoveltyFirestoreManager.kt |

All changes have direct evidence in the trace symbols. No optimizations were applied based solely on static code analysis.

# Appendix — Markdown Version

The following is the full report in Markdown format, suitable for inclusion in a README, wiki, or pull-request description.

```
# Threading Micro-optimizations Report – G45

**Project:** G45 · Sprint 4
**Author:** juan-jerez
**Date:** 2026-05-23
**Tool:** Android Simpleperf – `cpu-clock` + `sched:sched_switch`
**Trace file:** `cpu-simpleperf-G45.trace` (93 MB)
**Files changed:** 5

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## 1. Analysis Methodology

The trace was recorded with Simpleperf using two simultaneous events:
**cpu-clock** (CPU cycle samples every ~10 ms) and **sched:sched_switch**
(kernel-level thread context switches). The binary was inspected with `strings`
to extract method symbols present in sampled call stacks. Each appearance of a
symbol means that method was **actively executing on the CPU** during a sample.

A baseline was established: most ViewModels appear **2 times** in the symbol
set (e.g. `AgendaViewModel.`, `HomeViewModel.`). Symbols exceeding
that baseline were flagged as anomalies.

| Symbol | CPU samples | Interpretation |
|-----|-----|-----|
| `AgendaViewModel.` | 2x | Normal baseline |
| `HomeViewModel.` | 2x | Normal baseline |
| `PqrListViewModel.` | **6x** | ■ 3x above baseline |
| `TutorViewModel.` | **6x** | ■ 3x above baseline |
| `PqrViewModelFactory.create` | **6x** | ■ 3x above baseline |
| `listenToUserSessions$1.invoke` | **4x** | ■ concurrent with 2 others |
| `listenToReviews$1.invoke` | **4x** | ■ concurrent with 2 others |
| `listenToUnreadNovelties$1.invoke` | **4x** | ■ concurrent with 2 others |
| `flatMapLatest$1.invoke` (ReviewListVM) | **4x** | ■ 2 restarts per reload |

---

## 2. Finding 1 – Concurrent Fetches from ViewModel Recreation

> **Profiling evidence:** `PqrListViewModel.` → **6 CPU samples** (3x the
> 2-sample baseline). `PqrViewModelFactory.create` → **6 CPU samples**.

### Diagnosis

`PqrListViewModel` appears 6 times in the trace, indicating it was constructed
at least 3 times during the session. Each construction fires `fetchPqrs()` in
`init {}`. Without a cancellation guard, if the previous ViewModel was not
destroyed before a new one started, **both fetches run concurrently**: two
parallel Firestore reads for the same data, with two `_uiState` updates arriving
in non-deterministic order.
```

```
### Fix – `PqrListViewModel.kt`
```

```
```kotlin
```

```
// Before
```

```
fun fetchPqrs() {  
    val userId = AuthHolder.authRepo.getCurrentUserId() ?: return  
    viewModelScope.launch {  
        _uiState.update { it.copy(isLoading = true) }  
        // Firestore fetch...  
    }  
}
```

```
// After
```

```
private var fetchJob: Job? = null  
  
fun fetchPqrs() {  
    val userId = AuthHolder.authRepo.getCurrentUserId() ?: return  
    fetchJob?.cancel() // cancel in-flight fetch  
    fetchJob = viewModelScope.launch {  
        _uiState.update { it.copy(isLoading = true) }  
        // Firestore fetch...  
    }  
}  
```
```

**Effect:** Eliminates the race condition where two Firestore calls update  
`\_uiState` in non-deterministic order.

---

```
## 3. Finding 2 – Double-Emit Causing Redundant flatMapLatest Restarts
```

```
> Profiling evidence: `ReviewListViewModel$listenReviews$1$invokeSuspend`  
> `$$inlined$flatMapLatest$1.invoke` → 4 CPU samples. Each invocation  
> cancels and relaunches an inner coroutine.
```

```
### Diagnosis
```

`reloadReviews()` forced a re-emission by assigning `null` then the original  
value back to `\_dayFilter`. This produces **two distinct emissions** from  
`combine()`, each restarting the `flatMapLatest` inner coroutine – the first  
is cancelled mid-execution and discarded. With 2 reload events during the  
session, the lambda appears **4 times** instead of the expected 2.

```
### Fix – `ReviewListViewModel.kt`
```

```
```kotlin
```

```
// Before
```

```
fun reloadReviews() {  
    _uiState.update { it.copy(error = "") }  
    val current = _dayFilter.value  
    _dayFilter.value = null // emit 1 → restarts flatMapLatest  
    _dayFilter.value = current // emit 2 → restarts flatMapLatest  
}
```

```
// After
private val _refreshTrigger = MutableSharedFlow(extraBufferCapacity = 1)

fun reloadReviews() {
    _uiState.update { it.copy(error = "") }
    _refreshTrigger.tryEmit(Unit) // exactly 1 emit
}

// In listenReviews():
combine(
    _isTutor, _dayFilter,
    _refreshTrigger.onStart { emit(Unit) }
) { t, d, _ -> Pair(t, d) }
    .flatMapLatest { ... }
    ...

**Effect:** Each `reloadReviews()` call triggers exactly one flatMapLatest invocation instead of two. Cancelled mid-flight coroutines are eliminated.
```

---

#### ## 4. Finding 3 – Three Firestore Listeners Executing Simultaneously

```
> Profiling evidence: Three Firestore callback lambdas appear at the same
> frequency (4x each) within the same trace window.
```

#### ### Diagnosis

All three listeners are `callbackFlow` implementations. The Firestore SDK dispatches snapshots on its own thread pool. When all three receive an update simultaneously (e.g. initial snapshot on session start), their callbacks run in parallel. The default `callbackFlow` buffer is `Channel.BUFFERED` (64 slots), so if Firestore sends rapid snapshots while the collector is busy, they **accumulate in the buffer** and each triggers a redundant `\_uiState` update.

### Fix – `ReservationsFireStoreManager.kt`, `ProfileListsFireStoreManager.kt`, `NoveltyFirestoreManager.kt`

```
```kotlin
```

```
// Before
```

```
fun listenToUserSessions(...): Flow> = callbackFlow {
    // query + addSnapshotListener...
    awaitClose { registration.remove() }
}
```

```
// After
```

```
fun listenToUserSessions(...): Flow> = callbackFlow {
    // query + addSnapshotListener...
    awaitClose { registration.remove() }
}.conflate() // only latest snapshot delivered when collector is busy
...`
```

**Effect:** Intermediate Firestore snapshots are discarded when the collector is busy. The final state is always correct (most recent snapshot) with fewer CPU cycles and fewer redundant `\_uiState` updates.

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## ## 5. Change Summary

| # | Finding                                                  | Trace symbol                                                                        | Samples | Modified file                                                                                                      |
|---|----------------------------------------------------------|-------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------|
| 1 | Concurrent fetches from ViewModel recreation             | `PqrListViewModel.`                                                                 | 6×      | `PqrViewModelFactory.create` 6×                                                                                    |
| 2 | Double-emit causing redundant flatMapLatest restarts     | `listenReviews\$flatMapLatest\$1.invoke`                                            | 4×      | `ReviewListViewModel.kt`                                                                                           |
| 3 | 3 simultaneous Firestore listeners with buffered channel | `listenToUserSessions\$1`,<br>`listenToReviews\$1`,<br>`listenToUnreadNovelties\$1` | (4×     | each   4×                                                                                                          |
|   |                                                          |                                                                                     |         | each  <br>`ReservationsFirestoreManager.kt`,<br>`ProfileListsFirestoreManager.kt`,<br>`NoveltyFirestoreManager.kt` |

> All changes have direct evidence in the trace symbols.

> No optimizations were applied based solely on static code analysis.